AI 101 for Mechanical Engineers · Module 7

A minimal, honest regression workflow

Lesson 7.3

What you will learn

> Follow a regression workflow that separates fitting, model selection and final evaluation, then report the evidence and limitations of the result.

Lesson overview

Module 7 · Lesson 3

A minimal, honest regression workflow

AI 101 for Mechanical Engineers

1 Define the prediction problem — State the output, its unit and the inputs available at prediction time. Check the data source, and set aside relevant unseen examples before fitting.

2 Keep three stages separate

1 Training
Fit model parameters and learned preprocessing on training data.

▼

2 Validation
Compare model choices and input features.

▼

3 Final test
Evaluate the locked model after selection is complete.

Apply stored preprocessing to validation and test inputs. Do not refit it there.

3 Compare fairly and choose suitable splits

Compare with a training-mean baseline and, where available, a physical baseline on the same evaluation examples.

MAE and RMSE	R^2
in target units	dimensionless, relative

Match the split to the task

- ✓ Respect time, machine and event relationships.
- ✓ The pump lab uses independent synthetic designs, so a random split suits that example.
- ✓ Forecasting needs a split that respects time.

There is no universally correct split percentage.

4 Report the result and its limits

A regression report states

- ✓ The predicted quantity and its units
- ✓ Data provenance and split strategy
- ✓ Held-out metrics and fair baselines
- ✓ The operating range and limits on transfer

Provenance: where the data came from, including whether they are synthetic or measured.

Synthetic teaching data do not validate equipment safety. Evidence from one laboratory pump does not show performance on every pump.

Key takeaway

Once final-test results guide changes, those data become development data.
New independent evidence is then needed for an unbiased final assessment.

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Define the prediction problem first

- State the output and inputs available at prediction time
- Inspect the measurements, their units and their provenance
- Set aside relevant unseen examples before any fitting
- Provenance includes whether examples are synthetic or measured [38]

Three stages, three distinct jobs



- Training: fit parameters and learned preprocessing
- Validation: compare model configurations and input features
- Final test: evaluate the locked model after selection

Test fairly and report metrics in units

- Apply stored preprocessing to test examples without refitting
- Compare with training-mean and, where available, physical baselines
- Report MAE and RMSE in the target units
- R^2 is relative and dimensionless, not an error in W or K

No split percentage is universally correct

- No universally correct train-validation-test percentage [39]
- Respect time, machine, event and class structure
- Cross-validation can support development with limited data
- Once test results guide changes, new independent evidence is needed

Lab 2 demonstrates procedure, not equipment safety

What it covers

- Independent synthetic pump operating designs
- Ordinary inputs versus hydraulic power feature
- Random split suits this teaching problem

What it does not show

- Physical equipment safety
- Automatic transfer to every pump
- Forecasting, which needs a time-respecting split

Final-test metrics are reported only after validation selection.

A regression report states four things

1. The predicted quantity and its units
2. Data provenance and the split strategy
3. Held-out metrics and fair baseline performance
4. Evaluated operating range and limitations on transfer

Say what the evaluation supports

- A high R^2 alone cannot support a global safety claim
- An overall average can hide poor performance in part of the range

CHECK YOUR UNDERSTANDING

What is the difference between validation and the locked final test?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What is the difference between validation and the locked final test?

> Validation supports choices among models or features. The locked final test estimates performance after those choices are complete.

CHECK YOUR UNDERSTANDING

State two limitations to include alongside a regression score.

Answer before moving on.

CHECK YOUR UNDERSTANDING

State two limitations to include alongside a regression score.

> State whether the data are synthetic or measured and identify the operating range or machine population evaluated. Evidence from one laboratory pump does not establish performance on every pump.

Key takeaways

- Define available inputs and inspect measurement evidence
- Separate training, validation and final testing
- Compare against baselines and retain metric units
- Match the split strategy to the learning problem
- Report provenance, domain limits and the scope of the evidence

References

[38] T. Gebru et al., “Datasheets for datasets,” Commun. ACM, vol. 64, no. 12, pp. 86–92, 2021. doi: 10.1145/3458723

[39] S. Arlot and A. Celisse, “A survey of cross-validation procedures for model selection,” Statist. Surv., vol. 4, pp. 40–79, 2010. doi: 10.1214/09-SS054

Numbers match the course reference list. The written lesson is the authoritative text.